OS LAB-3
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1) SJF Non-Preemptive:

```
#include <stdio.h>
int main() {
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int pid[n], at[n], bt[n], ct[n], tat[n], wt[n], completed[n];
    float avg_tat = 0, avg_wt = 0;

    for (int i = 0; i < n; i++) {
        pid[i] = i + 1;
        printf("P%d AT: ", i + 1);
        scanf("%d", &at[i]);
        printf("P%d BT: ", i + 1);
        scanf("%d", &bt[i]);
        completed[i] = 0;
    }

    int time = 0, count = 0;
    while (count < n) {
        int pos = -1;
        for (int i = 0; i < n; i++) {
            if (at[i] <= time && completed[i] == 0) {
                if (pos == -1 || bt[i] < bt[pos]) {
                    pos = i;
                }
            }
        }

        if (pos == -1) {
            time++;
        } else {
            ct[pos] = time + bt[pos];
            tat[pos] = ct[pos] - at[pos];
            wt[pos] = tat[pos] - bt[pos];

            avg_tat += tat[pos];
            avg_wt += wt[pos];

            time = ct[pos];
            completed[pos] = 1;
            count++;
        }
    }
```

```c
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n", pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
    }
    printf("\nAverage TAT: %.2f", avg_tat / n);
    printf("\nAverage WT: %.2f\n", avg_wt / n);

    return 0;
}
```

OUTPUT:

```
[sumantshridhar@Sumants-MacBook-Pro OSLab % gcc SJF_NonPreemptive.c -o SJF_NonPreemptive
 sumantshridhar@Sumants-MacBook-Pro OSLab % ./SJF_NonPreemptive
[Enter number of processes: 5
 P1 AT: 0
 P1 BT: 3
 P2 AT: 2
 P2 BT: 2
 P3 AT: 3
 P3 BT: 5
 P4 AT: 4
 P4 BT: 4
 P5 AT: 6
 P5 BT: 1

 PID     AT      BT      CT      TAT     WT
 P1      0       3       3       3       0
 P2      2       2       5       3       1
 P3      3       5       15      12      7
 P4      4       4       9       5       1
 P5      6       1       10      4       3

 Average TAT: 5.40
 Average WT: 2.40
```

2) SJF Preemptive

```
#include <stdio.h>

int main(){
    int n,i,time,done = 0;
    float avg_wt = 0, avg_tat = 0;

    printf("Enter no. of processes: ");
    scanf("%d",&n);

    int pid[n],at[n],bt[n],rem[n],wt[n],tat[n],ct[n],completed[n];

    printf("Enter AT and BT:\n");
    for(i=0;i<n;i++){
        pid[i] = i+1;
        printf("P%d AT: ",i+1);
        scanf("%d",&at[i]);
        printf("P%d BT: ",i+1);
        scanf("%d",&bt[i]);

        rem[i] = bt[i];
        completed[i] = 0;
    }

    for(time = 0; done<n; time++){
        int shortest = -1;

        for(i=0;i<n;i++){
            if(!completed[i] && at[i]<= time)
                if(shortest==-1 || rem[i] < rem[shortest]) //no process selected yet or new shortest time
                    shortest = i;
        }
        if(shortest == -1) continue; //all arrived/completed= no process available
        rem[shortest]--;

        if(rem[shortest]==0){ //process completed
            completed[shortest] = 1;
            ct[shortest]=time+1;
            done++; //done should reach no of processes
        }
    }

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
    for(i=0;i<n;i++){
        tat[i] = ct[i] - at[i];
```

```
        wt[i] = tat[i] - bt[i];

        avg_tat += tat[i];
        avg_wt += wt[i];

        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
    }
    printf("\nAverage TAT: %.2f",avg_tat/n);
    printf("\nAverage WT: %.2f",avg_wt/n);

    return 0;
}
```

```
sumantshridhar@Sumants-MacBook-Pro OSLab % gcc SJF_Preemptive.c -o SJF_Preemptive
sumantshridhar@Sumants-MacBook-Pro OSLab % ./SJF_Preemptive
Enter no. of processes: 5
Enter AT and BT:
P1 AT: 0
P1 BT: 3
P2 AT: 2
P2 BT: 2
P3 AT: 3
P3 BT: 5
P4 AT: 4
P4 BT: 4
P5 AT: 6
P5 BT: 1

PID     AT      BT      CT      TAT     WT
P1      0       3       3       3       0
P2      2       2       5       3       1
P3      3       5       15      12      7
P4      4       4       10      6       2
P5      6       1       7       1       0

Average TAT: 5.00
Average WT: 2.00%
```

3) Priority Non-Preemptive

```c
#include <stdio.h>

int main() {
    int n;
    printf("Enter no. of processes: ");
    scanf("%d", &n);

    int pid[n], at[n], bt[n], pr[n], ct[n], tat[n], wt[n], completed[n];
    float avg_tat = 0, avg_wt = 0;

    printf("Enter AT, BT, and Priority:\n");
    for (int i = 0; i < n; i++) {
        pid[i] = i + 1;
        printf("P%d AT: ", i + 1);
        scanf("%d", &at[i]);
        printf("P%d BT: ", i + 1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i + 1);
        scanf("%d", &pr[i]);
        completed[i] = 0;
    }

    int time = 0, done = 0;
    while (done < n) {
        int pos = -1;
        for (int i = 0; i < n; i++) {
            if (!completed[i] && at[i] <= time) {
                if (pos == -1 || pr[i] < pr[pos]) {
                    pos = i;
                }
            }
        }

        if (pos == -1) {
            time++; // CPU idle
            continue;
        }

        ct[pos] = time + bt[pos];
        tat[pos] = ct[pos] - at[pos];
        wt[pos] = tat[pos] - bt[pos];

        avg_tat += tat[pos];
        avg_wt += wt[pos];
```

```
        time = ct[pos];
        completed[pos] = 1;
        done++;
    }

    printf("\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
               pid[i], at[i], bt[i], pr[i], ct[i], tat[i], wt[i]);
    }

    printf("\nAverage TAT: %.2f", avg_tat / n);
    printf("\nAverage WT: %.2f\n", avg_wt / n);

    return 0;
}
```

```
sumantshridhar@Sumants-MacBook-Pro OSLab % gcc Priority_NonPreemptive.c -o Priority_NonPreemptive
sumantshridhar@Sumants-MacBook-Pro OSLab % ./Priority_NonPreemptive
Enter no. of processes: 5
Enter AT, BT, and Priority:
P1 AT: 0
P1 BT: 3
P1 Priority: 5
P2 AT: 2
P2 BT: 2
P2 Priority: 4
P3 AT: 3
P3 BT: 5
P3 Priority: 1
P4 AT: 4
P4 BT: 4
P4 Priority: 2
P5 AT: 6
P5 BT: 1
P5 Priority: 6

PID     AT      BT      PR      CT      TAT     WT
P1      0       3       5       3       3       0
P2      2       2       4       14      12      10
P3      3       5       1       8       5       0
P4      4       4       2       12      8       4
P5      6       1       6       15      9       8

Average TAT: 7.40
Average WT: 4.40
```

4) Priority Preemptive

```
#include <stdio.h>

int main(){
    int n,i,time,done = 0;
    float avg_wt = 0, avg_tat = 0;

    printf("Enter no. of processes: ");
    scanf("%d",&n);

    int pid[n],at[n],bt[n],pr[n],rem[n],wt[n],tat[n],ct[n],completed[n];

    printf("Enter AT,BT,and Priority:\n");
    for(i=0;i<n;i++){
        pid[i] = i+1;
        printf("P%d AT: ",i+1);
        scanf("%d",&at[i]);
        printf("P%d BT: ",i+1);
        scanf("%d",&bt[i]);
        printf("P%d Priority: ",i+1);
        scanf("%d",&pr[i]);

        rem[i] = bt[i];
        completed[i] = 0;
    }

    for(time = 0; done<n; time++){
        int best = -1;

        for(i=0;i<n;i++){
            if(!completed[i] && at[i]<= time) //incomplete and arrived
                if(best==-1 || pr[i] < pr[best]) //no process selected yet or found smtn with better
priority
                    best = i;
        }
        if(best == -1) continue;
        rem[best]--;

        if(rem[best]==0){
            completed[best] = 1;
            ct[best]=time+1;
            done++;
        }
    }
```

```c
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
    for(i=0;i<n;i++){
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        avg_tat += tat[i];
        avg_wt += wt[i];

        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);
    }
    printf("\nAverage TAT: %.2f",avg_tat/n);
    printf("\nAverage WT: %.2f",avg_wt/n);

    return 0;
}
```

```
[sumantshridhar@Sumants-MacBook-Pro OSLab % gcc Priority_Preemptive.c -o Priority_Preemptive
[sumantshridhar@Sumants-MacBook-Pro OSLab % ./Priority_Preemptive
Enter no. of processes: 5
Enter AT,BT,and Priority:
P1 AT: 0
P1 BT: 3
P1 Priority: 5
P2 AT: 2
P2 BT: 2
P2 Priority: 4
P3 AT: 3
P3 BT: 5
P3 Priority: 1
P4 AT: 4
P4 BT: 4
P4 Priority: 2
P5 AT: 6
P5 BT: 1
P5 Priority: 6

PID     AT      BT      CT      TAT     WT
P1      0       3       14      14      11
P2      2       2       13      11      9
P3      3       5       8       5       0
P4      4       4       12      8       4
P5      6       1       15      9       8

Average TAT: 9.40
Average WT: 6.40%
```